1 Explain 3 OOP principles.

=> The 3 OOP principles are :

① Encapsulation :
* Building data (variables) and methods that operate on the data into a single unit (class)
* Access to data is restricted using access modifiers
* Getters and setters are used to access private variables.

② Inheritance
* One class (child/subclass) can inherit fields and methods from another class
* Promotes code reusability.

Example :

```
class Animal {
    void sound {
    System.out.println("Animal makes a sound");
    }
}
class Dog extends Animal {
    void bark() {
    System.out.println("Dog barks");
    }
}
```

=> Dog can access both sound() and bark().

③ Polymorphism
Ability of one method to behave differently based on the object calling it.

Example:

```
class shape {
    void draw () {
        System.out.println ("Drawing a shape");
    }
}
class circle extends shape {
    void draw () {
        System.out.println ("Drawing a circle");
    }
}
```

2. Write a program to find the biggest among 3 numbers using ternary operator (syntax also).

=> The ternary operator in java is a shorthand for if-else statements. It allows you to evaluate a condition and return one of two values depending on whether the condition is true or false.

Syntax:

```
condition ? value_if_true :
value_if_false;
```

Program:

```
Public class Main {
    public static void main (String[] args) {
        int a=10, b=25, c=15;
        // using nested ternary operator
        int biggest = (a>b)? (a>c)? a:c) : ((b>c)?
            b:c);
        System.out.println ("The biggest number is :" +
            biggest);
    }
}
```

Output :

The biggest number is 25.

3. Explain primitive data types.

=> Java has 8 primitive data types :

① byte
Size : 1byte (8 bits)
Range : -128 to 127
Example : byte b = 100;

② short
Size : 2bytes
Range : -32,768 to 32,767.
Example : short s = 30000;

③ int
Size : 4bytes
Range : -2,147,483,648 to 2,147,483,647.
Example : int num = 100000;

④ long.
Size : 8bytes
Range : Very large (±9quintillion)
Example : long l = 10000000000L;

⑤ float.
Size : 4 bytes
Used for decimal numbers (single precision)
Example : float f = 3.14f;

⑥ double
Size : 8 bytes
Used for decimal numbers (double precision)
Example : double d = 3.14159265 3;

⑦ char
Size : 2 bytes
Stores a single character (Unicode)
Example : char letter = 'A';

⑧ boolean
Values : true or false.
Use : For logical conditions and control flow.

4. Explain the switch statement with the program

⇒ A switch statement in java is used to execute one block of code among many based on the value of a variable.

Syntax :

```
switch (expression) {
  case value1 :
    // code block
    break;
  case value2 :
    // code block
    break;
  default :
    // default code block
}
```

Example program:

```
public class SwitchExample {
    public static void main (String[] args) {
        int day = 3;
        String dayName;
        switch (day) {
            case 1:
                dayName = "Monday";
                break;
            case 2:
                dayName = "Tuesday";
                break;
            case 3:
                dayName = "Wednesday";
                break;
            case 4:
                dayName = "Thursday";
                break;
            case 5:
                dayName = "Friday";
                break;
            case 6:
                dayName = "Saturday";
                break;
            case 7:
                dayName = "Sunday";
                break;
            default:
                dayName = "Invalid day";
        }
        System.out.println("Day" + day + "is" + dayName);
    }
}
```

5 Explain 4 different types of if statements in java.

=> There are four main types of if statements used for decision making.

1. Simple if statement:
Executes a block of code if the condition is true

```
int age = 20;
if (age >= 18) {
System.out.println("You are an adult.");
}
```

Output:
You are an adult.

② if-else statement.
Executes one block if the condition is true, another if its false.

```
int number = 10;
if (number % 2 == 0) {
System.out.println("Even number.");
} else {
System.out.println("Odd number.");
}
```

Output
Even number

③ if-else if-else ladder

Checks multiple conditions in sequence

```
int marks = 75;
if (marks >= 90) {
System.out.println ("Grade A");
} else if (marks >= 75) {
System.out.println ("Grade B");
} else if (marks >= 60) {
System.out.println ("Grade C");
} else {
System.out.println ("Grade D");
}
```

Output

Grade B

(4) Nested if statement.

An if statement inside another if.

```
int num = 15;
if (num > 0) {
if (num % 3 == 0) {
System.out.println ("Positive and divisible by 3");
} else {
System.out.println ("Positive but not divisible by 3");
}
}
```

Output :

Positive and divisible by 3

6) Explain while and do while loop with example

Sure! In Java, both `while` and `do-while` loops are used for repeated execution of code **as long as a condition is true**, but they behave slightly differently.

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## 1. while loop

- **Condition is checked first.**
- **If false at the start, the loop body won't execute at all.**

Java

Copy code

```
public class WhileExample {
    public static void main(String[]
args) {
        int i = 1;

        while (i <= 5) {
            System.out.println("Count: "
+ i);
            i++;
        }
    }
}
```

**Output:**

Copy code

```
Count: 1
Count: 2
Count: 3
Count: 4
Count: 5
```

---

## 2. do-while loop

- **Loop body is executed first**, then the condition is checked.
- **It always runs at least once**, even if the condition is false.

Java

Copy code

```java
public class DoWhileExample {
    public static void main(String[]
args) {
        int i = 1;

        do {
            System.out.println("Count: "
+ i);
            i++;
        } while (i <= 5);
    }
}
```

## Output (same as above):

Copy code

```
Count: 1
Count: 2
Count: 3
Count: 4
Count: 5
```